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Experiment no: 1

Name of the Experiment: Determination of the modulus of rigidity of the element of wire by the method of oscillation.

Questions on Theory

*1) What is elasticity? [0.5]

Ans: Elasticity is the ability of a body to resist a distorting influence and return to its original size and shape when that influence or force is removed.

*2) Define 'modulus of rigidity'. [1]

Ans: The modulus of rigidity is the elastic coefficient when a shear force is applied resulting in lateral deformation. It gives us a measure of how ~~g~~ rigid a body is.

$$Y = \frac{FL}{Al}$$

*3) Draw the arrangement of the experiment. [0.5]

Ans:

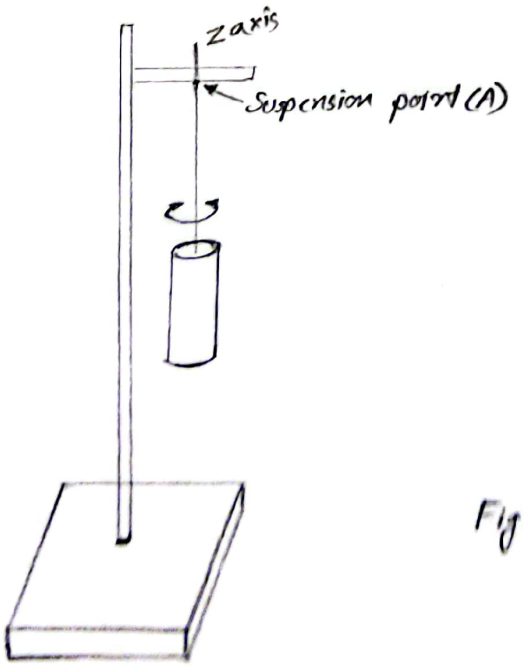

Fig: The arrangement of the experiment.

Elasticity

*4) Write down the working formula for this experiment. Mention what every variable represents. [1]

Ans: $\eta = \frac{8\pi I l}{T^2 r^4}$

I = moment of inertia

T = time period of circular oscillation

r = radius of wire

l = length of the wire

5) Suppose, we have two wires consisting of two <u>different materials</u>. Both of them are having same length and radius of cross section. Two identical cylinders are suspended vertically by means of them. The first one shows period of oscillation of 2s and the second one shows the period of oscillation of 4s. Which wire consists of the material of higher modulus of rigidity- the wire attached to the first cylinder or the wire attached to second cylinder? Why? [1]

Ans: $\eta_1 = \frac{8\pi I l}{T_1^2 r^4} = \frac{8\pi I l}{2^2 r^4}$ | $\eta_2 = \frac{8\pi I l}{T_2^2 r^4} = \frac{8\pi I l}{4^2 r^4}$

$T_1 = 2s$
$T_2 = 4s$

$\frac{\eta_1}{\eta_2} = \frac{8\pi I l}{2^2 r^4} \times \frac{4^2 r^4}{8\pi I l} = \frac{4^2}{2^2} = \frac{16}{4} = 4$

$\therefore \eta_1 = 4\eta_2$ $\boxed{\eta_1 > \eta_2}$

$\therefore$ modulus of rigidity of the wire attached to the first cylinder is highier.

6) Suppose, we have two wires consisting of <u>same material</u>. Both of them are having same radius of cross section. Two identical cylinders are suspended vertically by means of them. The length of the first wire is 1cm and that of the second wire is 2cm. Will the modulus of rigidity for different wires be different or same? If they are different then which one has higher modulus of rigidity? If they are same then why are they same? [1]

Ans: $\eta_1 = \frac{8\pi I l_1}{T^2 r^4} = \frac{8\pi I \times 1}{T^2 r^4}$ | $\eta_2 = \frac{8\pi I l_2}{T^2 r^4} = \frac{8\pi I \times 2}{T^2 r^4}$

$\frac{\eta_1}{\eta_2} = \frac{1}{2}$

$\Rightarrow 2\eta_1 = \eta_2$

$\boxed{\eta_2 > \eta_1}$ different rigidity

the 2cm wire has more rigidity. and the rigidity is not same.

Elasticity

বল প্রয়োগের ফলে বিকৃত হয়ে যাওয়ার ~~পর~~ কোন বস্তুর বল সরিয়ে নেয়ার পর আদি অবস্থায় ফিরে যেতে পারার সক্ষমতাকে স্থিতিস্থাপকতা বলা হয়।

Principles of Physics I (PHY111)

Lab

Experiment no: 1

Name of the experiment: Determination of the modulus of rigidity of the element of wire by the method of oscillation

Theory

Let's consider a cylinder of mass m and radius R which is suspended vertically by a thin circular wire from a point A, called the suspension point as shown in figure 1. The thin wire has negligible mass compare to the cylinder and has radius r and the modulus of rigidity η (if you are not familiar with this term please see appendix A given in the soft copy of this script which is available in the server). The wire is also aligned along the z axis.

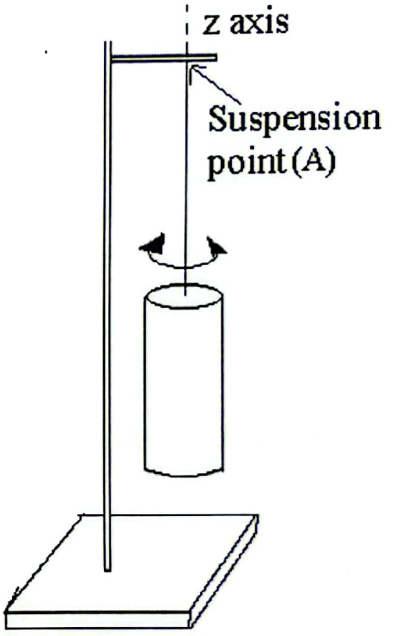

Figure 1: The arrangement of the experiment. A cylinder is suspended vertically by means of a wire.

If the cylinder is rotated through a small angle about z axis and then released, due to the rigidity modulus of the wire, the system (cylinder and the wire) will also oscillate rotationally. This is because of the Hooke's law. Let T be the time period of the circular oscillation. There is a relationship among η, T and the geometrical parameters of the wire. If l is the length of the wire, this relation is given by the following equation:

$$\eta = \frac{8\pi I l}{T^2 r^4} \qquad (1)$$

Where, I is the moment of inertia of the cylinder about z axis as shown in the diagram. Those who are interested to know how to derive this formula please see appendix B (given in the soft copy of this script which is available in the server).

If the mass of the cylinder is m and its radius is R, then its moment of inertia about its own axis is given by,

$$I = \frac{1}{2} m R^2 \qquad (2)$$

In this experiment you are going to measure the modulus of rigidity of the element of the wire by using equations (1) and (2).

Apparatus: Meter scale, slide calipers, screw gauge, stop watch, mass meter.

Modulus of rigidity

স্থিতিস্থাপক ~~সমা~~ সীমার মধ্যে বস্তুর কৃন্তন পীড়ন ও কৃন্তন বিকৃতির অনুপাত একটি ধ্রুব সংখ্যা। এই ধ্রুব সংখ্যাকে বস্তুর উপাদানের দৃঢ়তা গুণাঙ্ক বলে।

$$Y = \frac{\text{দৈর্ঘ্য পীড়ন}}{\text{দৈর্ঘ্য বিকৃতি}} = \frac{F/A}{l/L} = \frac{FL}{Al} = \frac{MgL}{\pi r^2 l}$$

Procedure:

1. Measure the length, l of the wire by using meter scale and record it in section A of the data sheet.

2. Measure the diameter of the wire three times at three different positions by using screw gauge. Compute their average. Find out the radius, r of the cross section of the wire. Use table 1- provided in the section B of data sheet, to record data of this step. To know how to use a screw gauge please see appendix C (given in the soft copy of this script that is available in the server).

3. Measure the diameter of the cylinder for three times at three different positions at the top, middle and bottom, by using slide calipers. Write down the data in table 2 of section B of the data sheet. Find out the radius, R of the cylinder. To know how to use a slide calipers please see appendix D (given in the soft copy of this script which is available in the server).

4. Record the mass of the cylinder in section D of the data sheet. You may possibly be given the value of the mass, m of the cylinder, if not, then measure the mass of the cylinder using the mass meter.

5. Rotate the cylinder through a small angle about its axis and release. It will oscillate rotationally. Record the time required to complete 20 oscillations by using stop watch and write it down in table 3 of section E. Repeat it for two more times and find their average. Then work out the period of oscillation, T by dividing total time by number of oscillations.

6. In the calculation part in the data sheet, compute the moment of inertia, I of the cylinder using equation (2) and using the data collected in step 3 and 5.

7. Putting the values found in steps 1, 2, 4 and 6 in equation (1) compute the modulus of rigidity of the wire

Read carefully and follow the following instructions:

- **Please READ the theory carefully, TAKE printout of the 'Questions on Theory' and ANSWER the questions in the specified space BEFORE you go to the lab class.**
- **To get full marks for the 'Questions on Theory' portion, you must answer ALL of these questions CORRECTLY and with PROPER UNDERSTANDING, BEFORE you go to the lab class. However, to ATTEND the lab class you are REQUIRED to answer AT LEAST the questions with asterisk mark.**
- **Write down your NAME, ID, THEORY SECTION, GROUP, DATE, EXPERIMENT NO AND NAME OF THE EXPERIMENT on the top of the first paper.**
- **If you face difficulties to understand the theory, please meet us BEFORE the lab class. However, you must read the theory first.**
- **DO NOT PLAGIARIZE. Plagiarism will bring ZERO marks in this WHOLE EXPERIMENT. Be sure that you have understood the questions and the answers what you have written, and all of these are your own works. You WILL BE asked questions on these tasks in the class. If you plagiarize for more than once, WHOLE lab marks will be ZERO.**
- **After entering the class, please submit this portion before you start the experiment.**